Road Risk Analysis Platform — Functional Module Documentation

Module Overview

Module	Objective	Main Role
AI Hazard Detection Module	Identify permanent road risks and infrastructure defects	Detect, locate, classify, and manage road hazards from recorded driving data
Predictive Cooperative Collision Prevention Module	Prevent potential collisions before they occur	Analyze vehicle interactions and provide proactive safety warnings
Community Reporting Module	Capture temporary road events reported by road users	Collect, validate, and distribute real-time road information from the community
Historical Risk Analysis Module	Identify dangerous areas based on historical patterns	Analyze accumulated driving and road data to identify risk zones and trends
Driver Behaviour Analytics Module	Analyze driving patterns and risky behaviours	Detect unsafe driving events and generate behavioural safety insights

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1. AI Hazard Detection Module

Sub-Module	Description	Functional Objective

Pothole Detection	Identification of damaged road surfaces and potholes	Detect recurring road defects that represent temporal risks
Traffic Sign Detection	Identification and classification of road signs	Monitor road signage availability, visibility, and compliancepermanent
Speed Bump Detection	Detection of speed control infrastructures	Identify speed bumps and their locations for driver awareness
Continuous Improvement and Retraining of AI Detection Models	Uses data from detections validated, rejected, or corrected by human reviewers to build and enrich training datasets for each detection model. Monitors historical results and model performance to identify when retraining is required.	Continuously improve the accuracy and reliability of all AI detection models by leveraging human validation feedback, accumulated data, and changes in model performance.
Parked/Roadside Vehicle Detection	Identification of vehicles creating possible road obstructions	Detect temporary physical obstacles affecting road safety
Hazard Location Management	Association of detected hazards with geographical locations	Ensure every detected risk can be located and communicated
Hazard Clustering	Grouping of nearby detections into road risk zones	Provide a consolidated view of repeated hazards in the same area
Detection Validation Workflow	Human review and approval process for detected hazards	Ensure reliability before hazards become available to users

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2. Predictive Cooperative Collision Prevention Module

Sub-Module	Description	Functional Objective
Telematics Data Collection	Periodic collection of telematics data from connected vehicles, including GPS position, speed, heading, and associated timestamps.	Provide the information required for real-time safety analysis
Live Vehicle Monitoring	Continuous monitoring of connected vehicles	Maintain awareness of active vehicles in the road environment
Road Context Analysis	Understanding vehicle position according to road structure	Differentiate meaningful interactions from simple geographic proximity
Trajectory Prediction	Estimation of future vehicle movements	Predict possible future conflicts between vehicles
Vehicle Conflict Detection	Identification of dangerous interactions between vehicles	Detect potential collision situations before they happen
Risk Evaluation Engine	Assessment of collision risk level	Determine the urgency of detected situations
Alert Management	Generation and management of driver warnings	Notify drivers when preventive action is required
Driver Safety Notification	Delivery of warnings through the user application	Help drivers react before a dangerous situation occurs
Road Intelligence Broker	Coordination of vehicle information and safety analysis	Manage relevant vehicle interactions within the road environment

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3. Community Reporting Module

Sub-Module	Description	Functional Objective

Road Event Reporting	User submission of road-related events	Allow drivers to contribute real-time road information
Traffic Condition Reporting	Reporting of congestion and traffic disruptions	Inform users about current traffic conditions
Police Presence Reporting	Reporting of checkpoints and enforcement activities	Warn drivers about temporary road controls
Accident Reporting	Reporting of road accidents	Provide immediate information about accident locations
Temporary Hazard Reporting	Reporting of temporary road dangers	Capture risks that cannot be detected permanently by AI
Road Closure Reporting	Reporting of blocked roads or unavailable routes	Help users avoid inaccessible roads
Weather Condition Reporting	Reporting of weather-related road risks	Communicate environmental conditions affecting safety
Roadside Assistance Requests	Reporting of assistance needs	Connect users with support services
Community Validation System	User confirmation or rejection of reports	Improve reliability through collective validation
Report Visibility Management	Control of report display according to geographical context	Show relevant information based on user location

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4. Historical Risk Analysis Module

Sub-Module	Description	Functional Objective

Driving Event Collection	Collection of historical driving-related events	Build a long-term understanding of road behaviour
Risk Data Aggregation	Grouping of historical safety information by geographical areas	Identify recurring risk patterns
Risk Score Calculation	Generation of safety indicators from historical data	Measure the risk level of locations and routes
Risk Zone Identification	Detection of areas with repeated safety problems	Identify dangerous locations requiring attention
Road Safety Heatmaps	Visualization of historical risk concentration	Provide a geographical view of safety trends
Historical Trend Analysis	Analysis of risk evolution over time	Understand whether road safety conditions improve or deteriorate
Route Safety Evaluation	Calculation of safety indicators for planned routes	Provide users with safer route recommendations
Risk Reporting Dashboard	Visualization of historical safety information	Support decision-making for authorities and operators

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5. Driver Behaviour Analytics Module

Sub-Module	Description	Functional Objective
Speeding Analysis	Detection of excessive speed events	Identify unsafe driving speed patterns
Harsh Braking Analysis	Detection of aggressive braking behaviour	Identify dangerous driving situations and possible road issues

Acceleration Analysis	Detection of sudden acceleration events	Identify risky driving behaviour
Cornering Analysis	Detection of unsafe turning behaviour	Identify dangerous driving patterns on curves and intersections
Driving Event Storage	Storage of behavioural events with location information	Maintain historical driving behaviour records
Behaviour Risk Scoring	Evaluation of driver behaviour severity	Generate safety indicators
Driver Behaviour Heatmaps	Visualization of risky driving concentration areas	Identify locations affected by unsafe behaviours
Driving Behaviour Statistics	Generation of behavioural reports and trends	Support fleet monitoring and safety improvement
High-Risk Zone Detection	Identification of areas with repeated unsafe behaviours	Reveal potential infrastructure or traffic problems

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Platform Integration View

Data Source	Related Module	Information Produced
Dashcam / Video Sources	AI Hazard Detection	Permanent road hazards
Connected Vehicles	Predictive Collision Prevention	Live vehicle safety information
Mobile Users	Community Reporting	Temporary road events
Vehicle Telemetry History	Historical Risk Analysis	Long-term road safety indicators
Driving Behaviour Data	Driver Behaviour Analytics	Driver and road behaviour insights

User Application Information Layer

Information Type	Source Module	User Benefit
Road Hazards	AI Hazard Detection	Awareness of permanent risks ahead
Temporary Events	Community Reporting	Awareness of current road conditions
Collision Warnings	Predictive Collision Prevention	Prevention of immediate dangers
Route Safety Score	Historical Risk Analysis	Selection of safer routes
Driving Insights	Driver Behaviour Analytics	Improvement of driving behaviour